Problem No: 04

Problem Name: Implement the Popular Image Enhancement Technique: Log Transformation & Power Log Transformation

Objective

- To understand and implement non-linear intensity transformation functions.
- To expand dark pixel values in an image using Log Transformation.
- To observe the effects of Gamma (γ) variations in Power-Law Transformation for image enhancement.

Theory:
Log Transformation: Log Transformation is defined as:

$$s = c\log(1 + r)$$

It maps a narrow range of low-intensity input values into a wider range of output levels. This technique is highly useful for enhancing details in the dark parts of an image.
Power-Law (Gamma) Transformation: Power-Law Transformation is defined as:

$$s = c \cdot r^{\gamma}$$

Depending on the value of γ, this transformation can either brighten or darken an image.

Program:

```
import cv2
import numpy as np
from matplotlib import pyplot as plt

image = cv2.imread('sm_pic.jpg', 0)
c = 255 / np.log(1 + np.max(image))
log_transformed = c * np.log(1 + image)
log_transformed = np.array(log_transformed, dtype=np.uint8)
gamma = 0.5
power_transformed = np.array(255 * (image / 255) ** gamma, dtype='uint8')
titles = ['Original Image',
          'Log Transformation',
          'Power-Law Transformation']
images = [image,
          log_transformed,
          power_transformed]
plt.figure(figsize=(12, 5))
for i in range(3):
    plt.subplot(1, 3, i+1)
    plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
plt.tight_layout()
plt.show()
```

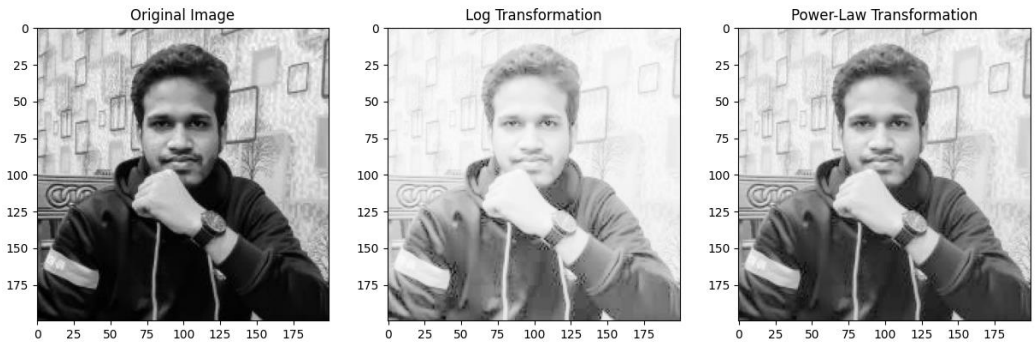
Output:

```
Log Transformed Image Matrix:

[[248 248 246 ... 237 237 242]
 [250 238 231 ... 247 248 245]
 [245 236 237 ... 245 246 246]
 ...
 [ 64  51  51 ... 119 122 126]
 [129 129 122 ... 111 115 115]
 [168 166 161 ... 111 107 111]]
```

```
Power-Law Transformed Image Matrix:

[[229 230 224 ... 204 204 214]
 [234 206 191 ... 226 229 222]
 [221 202 204 ... 221 224 223]
 ...
 [ 27  22  22 ...  55  57  59]
 [ 61  61  57 ...  50  52  52]
 [ 95  94  88 ...  50  47  50]]
```



Result & Discussion: Log Transformation brightened the dark regions and revealed hidden details, though it slightly reduced image contrast. In Power-Law Transformation, $\gamma = 0.5$brightened the image, while $\gamma = 1.5$and 2.2made the image darker with higher contrast. These methods effectively improved underexposed and overexposed images.

Analysis: Log Transformation expands low-intensity pixels, making dark areas clearer. Power-Law Transformation is more flexible because different γvalues control image brightness and contrast. Both are point processing techniques since each output pixel depends only on its original value.

Opinion: Through this experiment, I learned how Log and Power-Law transformations enhance image quality. I understood that gamma correction provides better control over brightness and contrast, which is useful in medical imaging, photography, and satellite imaging.